

```
9. apple.ta.rsi(length=14, append=True) # RSI14
10. apple.ta.bbands(length=20, append=True) # Bollinger Bands
11. apple.ta.macd(fast=12, slow=26, signal=9, append=True) # MACD
12. apple.ta.stoch(append=True) # Stochastic
13.
```

The pandas-ta library offers:

- **Ease of use:** Simple, intuitive API
- **Pandas integration:** Works directly with pandas DataFrames
- **Wide coverage:** Implements most common technical indicators
- **Easy installation:** Pure Python package with fewer dependencies

For beginners, pandas-ta provides an excellent balance of functionality and ease of use.

Custom Implementation: Building Your Own Indicators

While libraries handle standard indicators well, understanding how to implement them yourself gives you the flexibility to create custom indicators or modify existing ones:

```
1. def average_true_range(data, window=14):
2.     """Calculate Average True Range (ATR) - a volatility indicator"""
3.     high = data['High']
4.     low = data['Low']
5.     close = data['Close'].shift(1) # Previous close
6.
7.     # True Range is the greatest of:
8.     # 1. Current high minus current low
9.     # 2. Absolute value of current high minus previous close
10.    # 3. Absolute value of current low minus previous close
11.    tr1 = high - low
```